10.5 (a) For EM wave with frequency ω, we have $\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} = i\omega\vec{B}$. It is well-known that the solution to the curl equation $\nabla\times\vec{a} = \vec{b}$ can be written in a symmetric form as $\vec{a} = \frac{1}{2}\vec{b}\times\vec{r}$. Therefore, $\vec{E} = \frac{1}{2}i\omega\vec{B}\times\vec{r}$ and the induced current is $\vec{J} = \sigma\vec{E} = \frac{1}{2}i\omega\sigma\vec{B}\times\vec{r}$. The magnetic dipole moment for this current is given by

$$\vec{m} = \frac{1}{2}\int\vec{r}\times\vec{J}\,d^3x = \frac{i\omega\sigma}{4}\int\left(\vec{r}\times(\vec{B}\times\vec{r})\right)d^3x = \frac{i\omega\sigma}{4}\int\left[r^2\vec{B} - \vec{r}(\vec{r}\cdot\vec{B})\right]d^3x$$

The second term will have only non-zero component in the $\vec{B}$-direction. Thus,

$$\vec{m} = \frac{i\omega\sigma}{4}\int\vec{B}r^2\sin^2\theta\,d^3x = \frac{i\omega\sigma}{4}\vec{B}\cdot 2\pi\int_0^R r^4\,dr\int_{-1}^{1}\sin^2\theta\,d(\cos\theta)$$

$$= \frac{i\omega\sigma}{4}\vec{B}\cdot 2\pi\cdot\frac{R^5}{5}\cdot\frac{4}{3} = \frac{2i\pi\omega\sigma}{15}R^5\vec{B}$$

This is equivalent to the expression in the book, as $Z_0/\mu_0 = 1/\sqrt{\mu_0\varepsilon_0} = c$ and $\omega = ck$

(b) In the presence of both electric and magnetic dipole moments, the scattered electric field is

$$\vec{E}_{sc} = \frac{e^{ikr}}{r}\frac{k^2}{4\pi\varepsilon_0}\left[4\pi\varepsilon_0\frac{\varepsilon_r - 1}{\varepsilon_r + 2}E_0R^3(\vec{n}\times\vec{\varepsilon}_0)\times\vec{n} - \frac{i4\pi\sigma Z_0}{k\mu_0}(kR)^2\frac{R^3}{30}\frac{E_0}{c^2}\vec{n}\times(\vec{n}_0\times\vec{\varepsilon}_0)\right],$$

where we have used the fact that $\vec{B} = \frac{1}{c}\vec{n}_0\times\vec{E}$. Therefore,

$$\vec{f}(\vec{k},\vec{k}_0) = k^2\left[\frac{\varepsilon_r - 1}{\varepsilon_r + 2}R^3(\vec{n}\times\vec{\varepsilon}_0)\times\vec{n} + \frac{iZ_0\sigma}{k}(kR)^2\frac{R^3}{30}(\vec{n}_0\times\vec{\varepsilon}_0)\times\vec{n}\right]$$

Compared to Prob. 10.4, we can see that

$$\mathrm{Im}\left[\vec{\varepsilon}_0^{\,*}\cdot\vec{f}(\vec{k}=\vec{k}_0)\right] = \mathrm{Im}\left[\left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2}\right)k^2R^3 + iZ_0\sigma k(kR)^2\frac{R^3}{30}\right]|\vec{\varepsilon}_0\times\vec{n}_0|^2$$

$$= k^2R^3\frac{3Z_0\sigma/k}{(\varepsilon_r+2)^2 + (Z_0\sigma/k)^2} + Z_0\sigma k(kR)^2\frac{R^3}{30}.$$

The total cross section is given by the optical theorem,

$$\sigma_t = \frac{4\pi}{k}\mathrm{Im}\left[\vec{\varepsilon}_0^{\,*}\cdot\vec{f}(\vec{k}=\vec{k}_0)\right] = 4\pi\left[R^3\frac{3Z_0\sigma}{(\varepsilon_r+2)^2 + (Z_0\sigma/k)^2} + Z_0\sigma(kR)^2\frac{R^3}{30}\right]$$

$$= 12\pi R^2(RZ_0\sigma)\left[\frac{1}{(\varepsilon_r+2)^2 + (Z_0\sigma/k)^2} + \frac{(kR)^2}{90}\right].$$